

GoalOS Mission OS: The Equations of Proof-to-Action Work

Set the objective. GoalOS runs until proof is done.

AI creates output. GoalOS creates proof. A report informs; a governed decision state can be inspected, defended, acted on, and reused.

Mission value

$$V_{mission} = \frac{I_{decision} P_{integrity} A_{actionability} R_{reuse}}{C_{cost} + R_{risk} + L_{latency} + D_{proof\ debt}}$$

AI work

$$W_{AI} = O \times P \times V \times S \times R,$$
$$P = 0 \Rightarrow W_{trusted} = 0.$$

Acceptance

$$Accept(m) = 1 \iff ProofBundle(m) \wedge ValidatorClear(m)$$
$$\wedge SourceProvenance(m) \wedge RiskLedger(m)$$
$$\wedge ActionGraph(m) \wedge ClaimBoundaryPass(m).$$

Until-done loop

$$\Sigma_{t+1} = Repair(Generate(\Sigma_t), Q_t, B_t),$$
$$DONE = 1 \iff RequiredArtifactsComplete$$
$$\wedge EvidenceComplete \wedge BoundaryPass$$
$$\wedge QAReportPass \wedge HumanReviewReady.$$

Verifier Mesh

$$\mathcal{V}(m) = w_s S_{sources} + w_c C_{claims} + w_x X_{contradictions}$$
$$+ w_r R_{risk} + w_a A_{action} + w_b B_{boundary}.$$

Evidence Graph

$$EG_t = \{Missions, Claims, Sources, Proofs, Reviews,$$
$$Risks, Actions, Outcomes, Capabilities\}_{0:t}.$$

Moat

$$Moat(t) = \int_0^t (Q_{evidence} R_{reuse} X_{review} S_{settlement}) d\tau$$
$$- D_{proof\ debt} - K_{copyability}.$$

Operating doctrine

No proof, no settlement. No eval, no propagation. No
rollback, no release.

The deliverable is not a document. The deliverable is a governed decision state.